

Some ongoing efforts across the world...

1. Design of surface-mounted LED package using a ceramic sub-mount as the carrier and InGaN LED flip chips attached to a ceramic sub-mount
2. Improving wafer-level packaging to reduce cost; silicon-based wafer-level packaging offers high fabrication reliability, high yield, superior thermal management, integration of optical elements, and the direct integration of the driver IC in the package. The form factor can be further reduced using through-silicon vias, increasing the maximum amount of chips per wafer
3. Futuristic innovations in semiconductor, phosphorescent material and control
4. Optimisation of LED component platforms to help make LED lighting systems more cost-effective, efficient, robust, quicker to develop, higher in quality and with longer lifetimes
5. Use of reflectors instead of lenses to control optical properties in LED lighting systems, for various applications such as street lighting, back-lighting and adjustable office lighting
6. Overcoming the limitations in the design of plastic optical elements, which are cheaper, lighter and more versatile than glass optics
7. Reducing wastage due to over-illumination. Normally, the optics for LEDs produce circular or elliptical beams, which results in wastage when illuminating non-circular areas. New advances in optical design and mold tool manufacturing open the possibility of avoiding over-illumination by using optical components that make use of free-form optical surfaces
8. Combination of optical free-form surfaces with solid-state light sources for strongly asymmetric light distribution such as that required by many aesthetic applications and even street lighting, in a cost-effective way using methods like rapid prototyping
9. Design considerations for constant voltage, constant current and mixed-mode switching power supplies, including switching topologies and power factor correction configurations, in driving LED lighting systems
10. Sophisticated heat management, as required by the increase of luminous flux, usage of high-power LEDs, and thermal stress caused by extensive thermo-cycles
11. Use of multidimensional all-in-one boards to support flexible optical design and simple integration of intelligent light management electronics
12. Reducing fluctuations in the photometric characteristics within LEDs of one batch
13. Advance light measurement instrumentation required to detect the above-mentioned fluctuations (spectroradiometric light measurement techniques are being explored)
14. Modern spectrometric measurement systems and sophisticated algorithms for quick and precise calibration of systems consisting of a number of multi-chip LEDs—by measuring parameters like correlated colour temperature, colour coordinates or colour rendition indices, and automatically adjusting them in a reliable and predictive way.
15. Advanced 3D-pick-and-place machines that can be fully integrated in an automated assembly line; to explore new design opportunities for LEDs
16. New standards, norms and test procedures for the LED systems and technologies in terms of materials, drivers, system architecture, controls and photometric properties

Expanding applications

"LEDs in general are much more tailored to the specific requirements of the general lighting market than a couple of years before. Technical developments as well as a broad understanding of the customers' requirements together with industry adaptations can be seen in the latest SSL products. The new LEDs are dedicated to specific applications to gain the improvements of this technology," says Dr Fiebig. This, in many ways has impacted the type of applications that LEDs are being put to nowadays.

Solid-state LED lighting is already well-established in some market segments like architectural lighting, decoration/beautification, signage, automotive lighting, LCD backlights, wall washers, video walls, etc, and beginning to gain ground in other segments like outdoor facade lighting, professional shop lighting and residential lighting. The use of red, green and blue (RGB) combinations is expected to become increasingly popular in these segments due to the resulting flexibility of colour selection and the simplification of the optical aspects of lighting

designs. Independent market research institutes such as Strategies Unlimited have shown that the main market is in architectural lighting, which will be overtaken by retrofit lighting solutions. The residential sector, retail displays, shop lighting and outdoor lighting will support this growth.

If the improvements in performance and cost continue in the same trend, LED lamps might ram into the domestic market too and one day replace CFLs and even the ubiquitous 60W incandescent bulb. Dr Agrawal explains the possibility, "Ultra-high brightness (UHB) LED is becoming popular because of its high efficiency, small size, directional power property (incandescent, halogen or compact fluorescent bulbs give out omni-directional light) and long life. The operational life of current white LED lamps is 100,000 hours. This is eleven years of continuous operation, or 22 years of 50 per cent operation. The long operational life of an LED lamp is a big contrast to the average life of an incandescent bulb, which is approximately 5000 hours."

Using their intelligence intelligently

Due to their digital nature and ability to be controlled and managed, the SSL industry will be characterised above all by the development of intelligent solutions that adjust the amount of light according to demand and the particular situation. That is, 'tailor-made' light with optimum energy efficiency. Such intelligent solutions using presence or motion sensors will, in fact, bring out

Comparing the Three Common Forms of Lighting

	Bulb	CFL	LED
Wattage	40	13	7
Lumens	550	550	400
Efficacy (lm/W)	14	61	57
Life (hours)	5000	8000-10,000	100,000

Although CFL has a better efficacy than LED, other properties like direction, lifetime and size of LED makes it preferable in the domestic area

Courtesy: Dr Anita Agrawal